

BENGALURU PARKING ENFORCEMENT INTELLIGENCE

AI-Driven Illegal Parking Hotspot & Congestion Prioritization

FLIPKART GRID HACKATHON - DAY 1 & DAY 2 SUBMISSION

Focus Area: Predictive Patrol Pre-Positioning & Traffic Congestion Mitigation

Submitted by: Flipkart Hackathon Prototype Team

Prototype Stage: Fully Validated Client-Side Data & Forecasting Engine

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1. Executive Summary

The Bengaluru Parking Enforcement Intelligence system is a next-generation dashboard designed to transform city parking enforcement from a reactive posture to a targeted, predictive operation. In rapid urban centers like Bengaluru, illegal parking is not just a by-law infraction; it is a primary driver of localized gridlock, travel delays, and public transit inefficiencies.

This prototype integrates raw parking enforcement records with geographical metadata harvested from OpenStreetMap (OSM) to prioritize locations based on their real impact on traffic flow. Furthermore, it incorporates a temporal forecasting layer to predict exactly WHEN a given hotspot will spike next, allowing dispatchers to position enforcement patrols 15 to 30 minutes ahead of expected violations.

Key Achievements & Technical Deliverables:

- Snapped 3,452 unique coordinates to OSM road network nodes in under 1 second using a custom Spatial Grid Index.
- Built a multi-factor Congestion Impact Score based on road narrowness, POI transit proximity, and peak overlap.
- Trained a time-series predictor on 241,411 historical logs, reducing prediction error (MAE) on a test set by 4.15%.
- Created a flat-record frontend scanner running Set-based filters over 162k database cells in ~15ms.
- Implemented multi-select filters in the Control Panel and synced them to ranking lists and Leaflet map layers.

2. Prototype Architecture & Development Process

The prototype is structured as a standalone, zero-server frontend client utilizing a pre-compiled flat-file database. This structure ensures maximum portability, instantaneous loading speed, and eliminates API gateway latency during judge evaluations. The pipeline is divided into three layers:

2.1 Sourcing and Snapping Pipeline (OSM / Overpass API)

Raw violation records include GPS latitude and longitude coordinates. To evaluate congestion, these coordinates must snap to the nearest street segment. To achieve this without API rate limits, a python script grouped and deduplicated the coordinate centroids to 3,452 points and batched queries to multiple public OSM Overpass servers. The script fetched:

1. Road segments (highway type, number of lanes, speed limits, and road names).
2. Points of Interest (POIs) within 300m (metro entries, bus stations, markets, and malls).

A spatial grid index snapped each coordinate to its nearest road node. Flat service roads or narrow residential streets with 1 lane were assigned a higher narrowness penalty, while wide 3-lane trunk highways were penalized less.

2.2 Client-Side Optimized Database

A naive export of all raw coordinates and records would exceed 100MB, slowing down browser loading. To solve this, we aggregated records geographically to ~56-meter resolution grid cells (hotspots) with a minimum threshold of 5 violations. Detailed metadata was mapped to a dictionary in ``dashboard_data.js``.

The individual logs were then compressed into a highly compact, 7-column flat array ``FILTER_RECORDS`` containing indices pointing back to category maps (hour, weekday, vehicle type, violation type, status, and hotspot index). This compressed the file size to 10MB and enabled the browser to scan all records in memory.

3. Core Mathematical Algorithms & Techniques

3.1 Composite Hotspot Priority Score (Volume Mode)

In Volume Mode, hotspots are ranked by operational urgency. The composite priority score combines the raw frequency count, approval rate (to avoid acting on invalid citizen reports), and repeat offense rate (to prioritize chronic locations over isolated, single-day events):

$$\text{Priority Score} = 0.40 * \text{VolumeScore} + 0.30 * \text{ApprovalScore} + 0.30 * \text{RepeatOffenseScore}$$

where VolumeScore is the raw count scaled and capped at 500 violations, ApprovalScore is the approved reviewed share, and RepeatOffenseScore is the share of violations occurring outside the single peak-count day.

3.2 Congestion-Impact Score (Congestion Mode - Proxy Model)

To direct patrols to locations where parking violations cause the most severe traffic disruption, the dashboard implements a multiplicative proxy score:

$$\text{Congestion Impact} = \text{Density} * \text{Narrowness Penalty} * (1 + \text{Peak Hour Overlap}) * (1 + \text{POI Proximity Score})$$

- Density: Normalised parking violation density (resolution 56m).
- Narrowness Penalty: Scaled inverse of road capacity ($1 / (\text{highway_type_score} * \text{lanes})$). Ranges from 0.2 to 1.0.
- Peak Hour Overlap: Percentage of violations logged during daytime rush hours (8-11 AM and 5-8 PM IST).
- POI Proximity: Distance-decay sum of nearby metro stations, bus stops, and markets within 300 meters.

3.3 Time-Pattern Predictor (Spike Alerts)

Instead of relying on historical static counts, we trained a time-series regression model on the historical dataset. For each hotspot, we calculated a 7x24 weekly pattern matrix representing expected violation rates. Hotspots with data in fewer than 6 weeks were flagged as 'thinly_sampled' to warn patrols of low confidence.

The model was validated against a held-out test set (final 4 weeks), yielding a Mean Absolute Error (MAE) of 2.19 violations/hour, representing a 4.15% error reduction over the baseline overall average. The Patrol Spikes recommendation tab ranks hotspots by: Expected Next-Hour Rate * Congestion Score, allowing proactive positioning before congestion starts.

4. Dashboard Interface & Functional Layout

The frontend is built using Vanilla HTML, CSS, and JS, incorporating Leaflet.js for geographic maps, and Chart.js for data visualization. Below is the description of what each dashboard section does:

- Header Panel & KPIs:

Displays high-level statistics in real-time. Includes total violations filtered, active hotspots count, global enforcement confidence rate, and the review queue backlog share.

- Control Panel Sidebar:

Allows the user to select the scoring mode (Volume vs. Congestion), search for specific police stations or junctions, and filter the dataset by Vehicle Type, Violation Category, Time of Day, Day of Week, and Validation Status using custom checkbox-based multi-select dropdowns.

- Interactive Leaflet Map:

Visualizes the geographic distribution of violations. In Volume Mode, it renders a red/yellow flame heat gradient with neon red circles; in Congestion Mode, it swaps to a slate blue/white ice gradient with neon cyan circles. Clicking a marker loads predictions in the charts panel.

- Ranking Panel (Police Stations / Junctions / Patrol Spikes):

Ranks geographic areas dynamically. Swapping to 'Patrol Spikes' shows upcoming expected hotspot surges for the next hour based on the selected query time, assisting in dispatch scheduling.

- Time-Pattern Predictor Panel:

Displays a 24-hour predictions chart for a selected hotspot, jurisdiction, or junction. Highlighted bars show the planning window and the subsequent 3 hours. It alerts patrols if expected activity is high or if the data is thinly sampled.

- Hourly Temporal Profile Chart:

Displays the distribution of violations across 24 hours. A warning banner highlights anomalous peaks between 02:00 and 05:00, explaining that this is due to batch syncing of enforcement cameras and not actual traffic volume.

- Vehicle Mix & Validation Status Charts:

A vertical bar chart shows the breakdown of vehicles (scooters, cars, passenger autos, etc.) and a pie chart shows the share of approved, rejected, and pending review cases.

5. Data Cleaning & Calibration Disclosures

5.1 Database Reduction Flow

During data preparation, the records flowed through multiple filters to clean duplicate noise and focus strictly on hotspots:

- Raw CSV/Excel Rows: 298,450 records.
- After Deduplication: 293,078 records (5,372 duplicates removed).
- Hotspot Threshold (Count $\geq$ 5): 277,299 records (15,779 isolated violations excluded).
- Dashboard Records (Temporal predictions filter): 277,294 records (5 boundaries excluded).

5.2 Scientific and Operational Honesty

To maintain absolute credibility with judges and law enforcement officials, the dashboard contains prominent disclosures:

1. Proxy Scoring: We disclose that the congestion-impact score is a proxy metric modeled from road narrowness and transit proximity, not a live traffic speed forecast.
2. Anomalous Nighttime Peak: The chart warning clarifies that the 02:00 to 05:00 peak represents batch syncing of enforcement cameras and not actual traffic volume.
3. Low Confidence Alerts: Prediction plots show warnings when selected hotspots are thinly sampled to prevent dispatching patrols based on unstable data.

5.3 Calibration Roadmap (v2)

To scale the proxy model into a fully calibrated traffic-flow model, the next phase requires:

1. Google Directions API: Measure actual speed deltas on snapped road segments during peak hours.
2. Live Traffic Streams: Correlate active violation alerts with live traffic delays near hotspots.
3. Radar Calibration: Sync enforcement logs with road-side microwave traffic counters.